Mahesh Polisetty Data Scientist

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Summary

Results-driven Data Scientist with 4+ years of experience in building and deploying predictive models, pricing systems, and NLP applications across aviation, healthcare, and consumer electronics industries. Proficient in Python, SQL, and machine learning frameworks, delivering measurable business impact through advanced analytics and automation. Adept at working in Agile teams, integrating cloud and MLOps tools to scale data science solutions. Currently completing an MS in Data Engineering, reinforcing strong foundations in both data engineering and applied machine learnings.

Technical Skills

- **Programming Languages**: Python, SQL, R
- **Machine Learning & Deep Learning**: Scikit-learn, TensorFlow, Keras, XGBoost, Random Forest, Gradient Boosting, CNN, SVM, Autoencoders, Isolation Forest, Survival Analysis, BERT, Reinforcement Learning
- **Natural Language Processing (NLP)**: BERT, Text Preprocessing, Sentiment Analysis, Semantic Search
- **Data Processing & Wrangling**: Pandas, NumPy, OpenCV, Data Cleaning, Feature Engineering
- **Big Data & Cloud Platforms**: Azure DevOps, MLflow, Databricks, AWS (S3, SageMaker), Google Cloud Platform (GCP)
- **Data Visualization & Reporting**: Power BI, Tableau, Matplotlib, Seaborn
- **Model Deployment & MLOps**: Docker, Kubernetes, CI/CD Pipelines, Model Monitoring, Automated Retraining
- **Development Frameworks & Tools**: Jupyter Notebook, Git/GitHub, React, Supabase, Netlify Functions
- **Time Series & Statistical Modeling**: ARIMA, Cross-Validation, AUC-ROC, MAE, MSE, RMSE
- **Collaboration & Methodologies**: Agile, Scrum, Cross-Functional Teamwork, Documentation

Professional Experience

Data Scientist, Logitech — 01/2025 – Present | Remote, USA

- Developed advanced machine learning models to optimize pricing and promotion strategies by analyzing customer purchasing patterns, competitive pricing, and product lifecycle data—resulting in a 10% increase in revenue.
- Designed and implemented dynamic pricing algorithms and A/B testing frameworks that balanced profitability with customer loyalty, leading to a 5% improvement in average transaction value and a 6% reduction in return rates.
- Applied gradient boosting, survival analysis, and Bayesian techniques to forecast customer lifetime value and product upgrade propensity, enabling targeted marketing campaigns and personalized offers across consumer segments.
- Conducted in-depth analysis of user engagement, product usage, and churn behavior across key markets using Python (Pandas, Scikit-learn), SQL, and Power BI, ensuring >90% data integrity and identifying key retention drivers.
- Integrated LLMs (GPT-3.5-turbo, BERT) for automated customer feedback summarization and response generation; fine-tuned DistilBERT for complaint classification and co-developed an agent-assist AI tool, improving support efficiency by 25%.
- Experience with AI/Generative AI, including LLMs, Retrieval-Augmented Generation (RAG), and prompt engineering/evaluation, to accelerate model development and enhance customer experience automation.
- Automated reporting pipelines and dashboards using SQL and Tableau, streamlining business insights delivery and facilitating strategic reviews with leadership and cross-functional stakeholders.
- Collaborated cross-functionally with product managers, UX designers, and engineering teams in Agile environments to translate business needs into scalable AI solutions, accelerating product innovation and enhancing Logitech's data-driven decision-making.

Data Scientist, Indigo — 01/2021 – 07/2023 | Gurugram, India

- Built end-to-end ETL pipelines using Apache Spark and Python to process large-scale flight and passenger datasets, ensuring high data quality and efficient ingestion for downstream analytics and modeling.
- Developed and deployed machine learning models such as XGBoost and Random Forest for flight scheduling and dynamic pricing optimization. Collaborated with revenue management and operations teams to integrate model outputs into business processes.
- Conducted rigorous model training and evaluation, including hyperparameter tuning, cross-validation, and performance measurement using AUC-ROC and precision-recall metrics. Utilized Python libraries like pandas, NumPy, scikit-learn, TensorFlow, and PyTorch to enhance predictive accuracy.
- Designed and implemented anomaly detection systems using Isolation Forest and Autoencoder architectures to identify irregularities in customer feedback related to onboarding services. Integrated these models into real-time monitoring dashboards for operational teams.
- Applied BERT-based NLP models to analyze customer feedback and social media sentiment, extracting actionable insights to guide improvements in passenger experience and marketing strategies.
- Partnered with DevOps and IT teams to deploy ML models in production using MLflow, Azure DevOps, and Apache Airflow for automated retraining workflows. Documented all model assumptions and deployment processes to ensure scalability and reliability.
- Developed interactive dashboards using Plotly/Dash and Power BI to visualize pricing trends, operational KPIs, and model performance, enabling real-time decision-making for stakeholders.
- Executed A/B testing frameworks for pricing strategies and customer experience enhancements, ensuring statistically sound outcomes to support data-driven decisions.

Jr Data Scientist, MSD India — 01/2020 – 12/2020 | Hyderabad, India

- Processed multi-source healthcare data (EHR, genetics, lifestyle) using Python and data wrangling tools, ensuring data quality and privacy compliance for analysis in collaboration with cross-functional teams.
- Developed, tested, and evaluated machine learning models for personalized medication recommendations, applying supervised and reinforcement learning, incorporating clinician feedback, and documenting results to support iterative improvements and deployment.

Education

University of North Texas — Denton, TX, USA
Master of Science in Data Engineering — 08/2023 – 05/2025

Certificates

- Databricks Certified Data Analyst Associate
- JavaScript RAG Web Apps with LlamaIndex, deeplearning.ai
- ChatGPT Prompt Engineering for Developers, deeplearning.ai
- Large Language Models with Semantic Search, deeplearning.ai

Projects

- **Hospital voice assistant:** Developed a scalable voice assistant for hospital appointment scheduling using Gemini 2.0 LLM, React, Supabase, and Netlify Functions, enabling real-time speech interaction and improving patient accessibility with modular tools and webhook call summaries.
- **GDP Forecast:** Built a predictive model analyzing 50+ years of GDP data for top 20 economies using Random Forest and ARIMA in Jupyter, enhancing macroeconomic insights and financial planning, validated with MAE, MSE, and RMSE metrics.
- **Leaf detection:** Created a plant disease detection system using CNN and SVM on 75,000+ high-res leaf images with TensorFlow and OpenCV, automating early diagnosis, reducing manual inspections, and improving classification accuracy via deep learning and image processing.